Differential Photometry and Optical Light Curve of BL Lacertae

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Abstract

In this project, optical CCD observations of BL Lacertae were reduced and analyzed using differential photometry. The raw science frames were corrected using master bias, master dark, and master flat frames. Aperture photometry was then performed on BL Lac and three reference stars in the same field. The brightness of BL Lac was measured relative to the average brightness of the three reference stars in order to reduce atmospheric and instrumental effects. Light curves were produced for 24 August 2020 and 25 August 2020 separately, and also for the combined data set. The individual night light curves show smooth short term variability. The combined light curve shows a clear offset between the two nights. This may represent real source variability, but a systematic night to night offset cannot be ruled out without a dedicated comparison star stability test.

1 Introduction

BL Lacertae is a well known blazar and is a strongly variable active galactic nucleus. Optical variability studies of such sources are useful because they allow us to trace short term changes in the emission from the jet. The aim of this project is to construct the optical light curve of BL Lac from CCD observations and to understand how differential photometry helps in measuring brightness changes more reliably than raw counts alone.

If one directly compares raw source counts from image to image, the result is strongly affected by sky transparency, atmospheric extinction, detector response, and other observing conditions. Differential photometry avoids this problem by comparing the target with nearby stars in the same frame. Since the target and the comparison stars are observed under the same conditions, most common variations cancel when the brightness difference is taken. In this work, BL Lac was compared with three reference stars in the same field, and its light curve was constructed from those relative measurements.

2 Observations and Data

The data set contains CCD images taken in the R band on 24 August 2020 and 25 August 2020. The raw data also include calibration frames in the form of bias, dark, and flat

images. The available file inventory shows 5 flat frames on 20200819, 10 bias frames, 10 dark frames, and 16 science frames on 20200824, 10 bias frames, 10 dark frames, and 17 science frames on 20200825, and 5 additional flat frames on 20200830. All images have a size of 2048×2048, and the science observations use the same filter, `R_Astrodon_2018`. This consistency is important because it allows the calibration frames and science frames to be used together in a physically meaningful way.

3 CCD Reduction

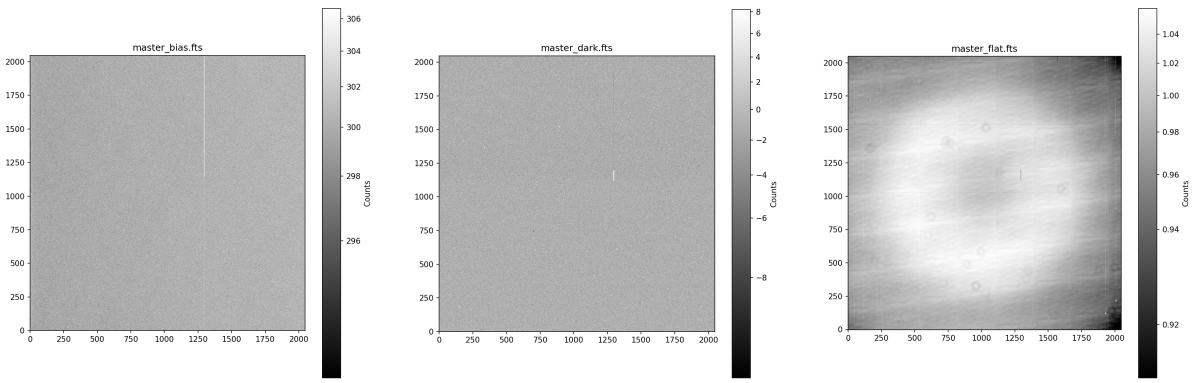
The raw science images cannot be used directly for photometry because they contain instrumental signatures. The first correction is the bias correction. A bias frame measures the electronic offset introduced during CCD readout. A master bias was created by averaging the individual bias frames. The second correction is the dark correction. A dark frame measures thermal signal produced inside the detector during the exposure. After subtracting the master bias from each dark frame, the dark frames were averaged to produce the master dark. The third correction is the flat field correction. Flat frames remove pixel to pixel sensitivity variations and large scale optical response variations such as vignetting or dust shadows. The flat frames were corrected using the master bias and master dark, then combined and normalized to produce the master flat.

The final calibrated science frame was computed using

$$I_{\text{cal}} = \frac{I_{\text{raw}} - B - D}{F},$$

where I_{raw} is the raw science image, B is the master bias, D is the master dark, and F is the normalized master flat. This is the standard CCD reduction chain used in the project instructions.

Figure 1 shows the calibration products used in the reduction. Figure 2 shows one representative calibrated science frame of BL Lac after all corrections.



(a) Master bias (b) Master dark (c) Master flat

Figure 1: Calibration products used for CCD reduction. The master bias corrects the readout offset, the master dark corrects thermal signal, and the master flat corrects detector and optical response variations.

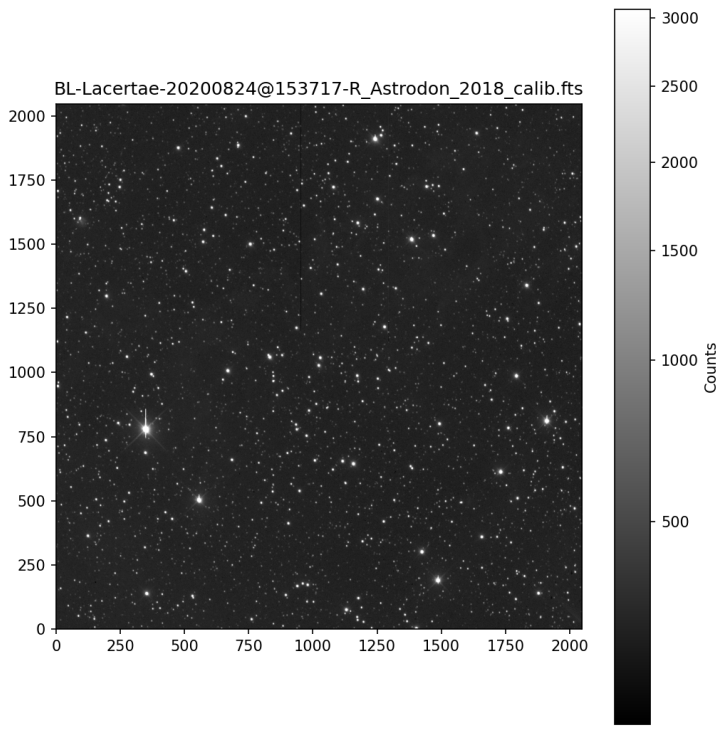


Figure 2: A representative calibrated science frame of BL Lac after bias, dark, and flat correction. This reduced image was used for aperture photometry.

4 Photometry Method

Photometry was carried out on the calibrated images using a circular aperture around each source and an annulus around the source for background estimation. The source positions and aperture definitions were taken from the DS9 region file prepared for BL Lac and the three reference stars. For each image, the counts inside the source aperture were summed. The sky contribution was estimated from the annulus and subtracted from the source counts to obtain the net flux,

$$F = F_{\text{source}} - F_{\text{sky}}.$$

The net flux was converted into instrumental magnitude using

$$m_{\text{inst}} = -2.5 \log_{10}(F).$$

This is not an absolute magnitude. It is only a detector based measure of brightness. The key step of the analysis is to compare the target with the reference stars in the same frame. In this project, the differential magnitude of BL Lac was calculated as

$$\Delta m = m_{\text{BL Lac}} - \langle m_{\text{ref}} \rangle,$$

where $\langle m_{\text{ref}} \rangle$ is the mean instrumental magnitude of the three reference stars.

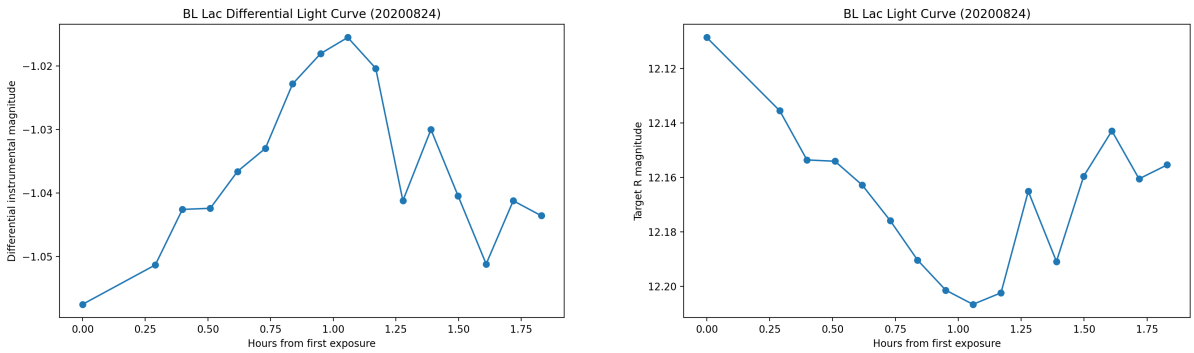
The intuition is simple. If the atmosphere changes slightly during the night, all stars in the same image become affected in nearly the same way. By subtracting the average magnitude of the reference stars from the target magnitude, most of the common atmospheric and instrumental effects cancel out. What remains is mainly the relative brightness change of BL Lac itself. In addition to the differential light curve, an approximate calibrated R magnitude of BL Lac was also obtained using the known magnitudes of the reference stars.

5 Results

The individual night light curves are shown in Figures 3 and 4. For each night, two curves are shown. The first is the differential light curve, and the second is the approximate calibrated R magnitude curve. The horizontal axis is plotted in hours from the first exposure of that night, which makes the short term behaviour easier to see.

On 24 August 2020, the differential light curve shows a gradual change during the first part of the observing run, followed by a decline with moderate scatter. The R magnitude plot shows the same trend, with the magnitude axis inverted in the usual astronomical convention. The behaviour is smooth enough to suggest that the reduction and photometry are functioning properly and that the observed variation is not dominated by random noise.

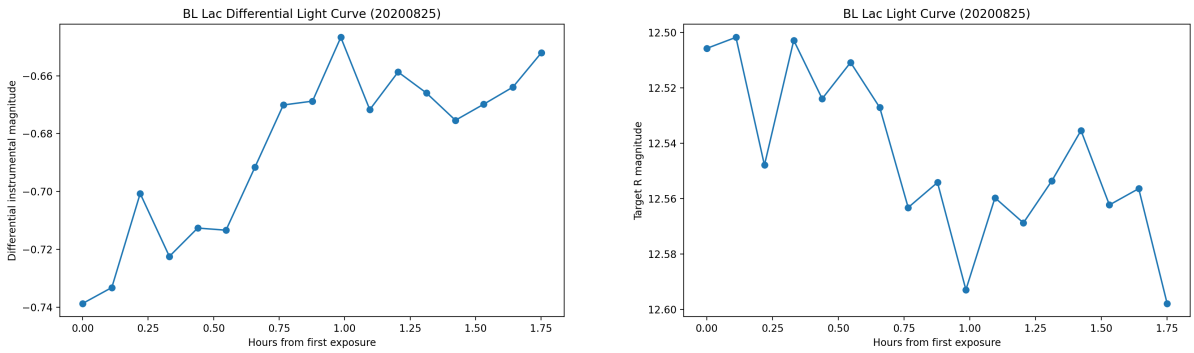
On 25 August 2020, the source again shows a smooth variation over the observing run. The differential light curve rises during the first part of the night and then stays at a relatively higher level with smaller fluctuations. The calibrated magnitude curve follows the same pattern. The fact that both nights show coherent variations rather than completely random scatter supports the conclusion that the photometric extraction is meaningful.



(a) Differential light curve

(b) Approximate R magnitude light curve

Figure 3: Light curves of BL Lac for 24 August 2020 plotted against hours from the first exposure.

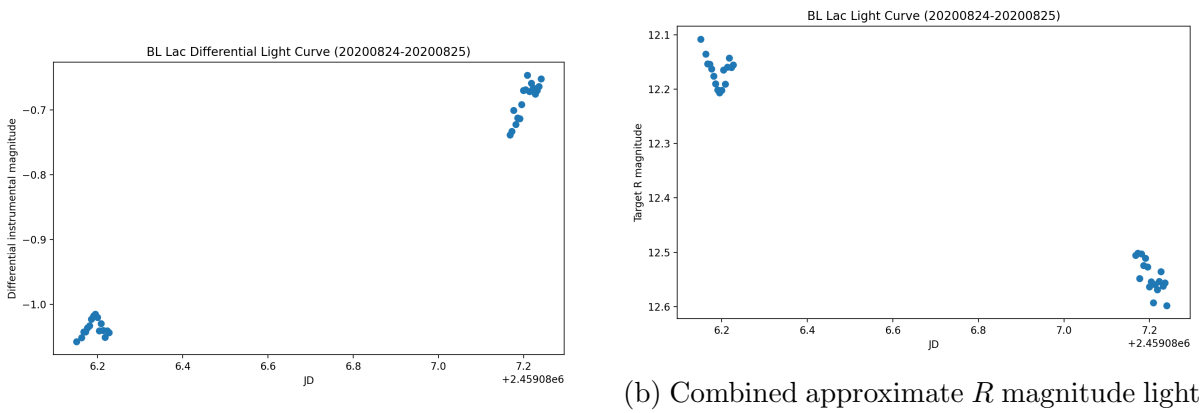


(a) Differential light curve

(b) Approximate R magnitude light curve

Figure 4: Light curves of BL Lac for 25 August 2020 plotted against hours from the first exposure.

The combined light curves for both nights are shown in Figure 5. The separation along the time axis is expected because the two groups of observations come from different Julian Dates, roughly one day apart. The more interesting feature is the vertical offset between the two nights. In the differential light curve, the second night lies noticeably above the first night. The calibrated magnitude curve also shows a shift, with BL Lac appearing fainter on 25 August than on 24 August. Since BL Lac is a variable blazar, this could represent real night to night variability. However, a systematic offset between nights cannot be ruled out without testing the stability of the comparison stars directly.



(a) Combined differential light curve

(b) Combined approximate R magnitude light curve

Figure 5: Combined light curves for 24 August 2020 and 25 August 2020. The offset between the two nights may be real, but it should be checked further using comparison star stability tests.

6 Discussion

The main result of this analysis is that BL Lac shows measurable short term optical variability on both observing nights. The reduction pipeline, which includes bias subtraction, dark subtraction, and flat field correction, produced calibrated science images suitable for photometry. The differential photometry method then removed most common observing

effects by comparing BL Lac with three stars in the same field. This is why the light curves are smoother and more reliable than any direct comparison of raw counts would have been.

The night to night offset in the combined light curve needs careful interpretation. Since BL Lac is known to vary, a real brightness change between the two nights is physically plausible. At the same time, the shift could be partly caused by a systematic photometric zero point difference between nights. The correct next step would be to construct light curves using only the comparison stars against one another. If those star to star curves remain flat, then the BL Lac offset is much more likely to be intrinsic to the source. If they also show an offset, then the night to night shift in BL Lac is at least partly systematic.

This analysis also does not include formal error bars derived from photon statistics, sky background uncertainty, and aperture placement. A more complete version of the work should propagate those uncertainties into the final light curves. Even without that step, the present analysis is enough to demonstrate the basic power of differential photometry for studying variable astronomical sources.

7 Conclusion

In this project, CCD observations of BL Lac were reduced and analyzed using standard astronomical image processing and differential photometry. Master bias, master dark, and master flat frames were produced and applied to the science data. Aperture photometry was then performed on BL Lac and three reference stars, and the light curve of BL Lac was constructed from the difference between the target magnitude and the average magnitude of the reference stars. The final results show clear intra night variability on both 24 August 2020 and 25 August 2020. The combined data also suggest a night to night change in brightness, although this result should be tested further through comparison star stability analysis.

Overall, the project shows that differential photometry is an effective and practical method for extracting source variability from CCD observations because it suppresses common atmospheric and instrumental effects and isolates the relative behaviour of the target.